Reach-Avoid Controllers Synthesis for Safety Critical Systems

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***Abstract*—In this article, we propose a framework for synthesizing reach-avoid controllers for deterministic systems modeled by ordinary differential equations and stochastic systems modeled by stochastic differential equations based on the notion of control guidance-barrier functions. We first consider deterministic systems and aim to synthesize a reach-avoid controller that modifies a nominal controller in a minimal way to enforce the reach-avoid objective. This objective ensures that the system enters a target set eventually while staying within an open safe set before the first target hitting time. Three control guidance-barrier functions and corresponding conditions for synthesizing reach-avoid controllers are developed. These conditions progressively weaken, facilitating optimal controller design. We then extend two of these functions for deterministic systems to stochastic systems to synthesize reach-avoid controllers in the probabilistic setting. Finally, we provide several examples to demonstrate the effectiveness of our proposed methods, using the semidefinite programming tool.**

***Index Terms*—Guidance-barrier functions, reach-avoid controllers, safety critical systems.**

I. Introduction

Cyber-physical systems are becoming ubiquitous due to rapid advances in computation, communication, and memory [14], and can be found in diverse application domains including automobiles, aviation, advanced manufacturing, and integrated medical devices [4]. Many of these systems are safety-critical, and thus, must satisfy safety requirements, which are typically formulated as forward invariance of a given safe set [2], on their dynamic trajectories in order to prevent economic harm and loss of life. The need for safety has motivated extensive research into verifying and synthesizing controllers to satisfy safety requirements.

Common techniques include Lyapunov and barrier methods [13], discrete approximations, and direct computations of reachable sets [1], [8], among others, have been developed for safety verification of safety-critical systems with given controllers. However, as the systems being verified get more complicated, the traditionally verified and tuning/redesigning ("design V") iteration procedure [7] becomes harder and more time-consuming. Consequently, there has been increasing interest in synthesizing controllers with provable safety guarantees in the design phase, and thus, many methods emerge, such as control barrier functions methods [2], Hamilton–Jacobi methods [16], moment-based optimization methods [10], and controlled invariant sets methods [5], [9]. Such methods are instrumental in removing bad controllers at an early state and avoiding the time-consuming "design V" iteration, consequently reducing software development expenses. Besides safety objectives, cyber-physical systems, such as autonomous vehicles, industrial robots, and chemical reactors often require simultaneous satisfaction of performance specifications. For instance, Ames et al. [3] considered the design of controllers enforcing objective of joint safety and stability for deterministc systems. Clark [6] proposed control policies for stochastic systems that ensure safety and stability with probability one by solving quadratic programs containing control barrier functions and stochastic control Lyapunov functions. Among the many provably correct controllers synthesis problems, there is a special one of synthesizing reach-avoid controllers. Reach-avoid controllers enforce both safety via avoiding a set of unsafe states and reachability via reaching a desired target set. Consequently, they can address many important engineering problems, such as collision avoidance [11] as well as target surveillance, and thus, have turned out to be of fundamental importance in engineering.

In this article, we address the problem of synthesizing reach-avoid controllers for continuous-time systems modeled by ordinary and stochastic differential equations. A reach-avoid controller is designed to ensure that the system, starting from a safe set, will eventually enter a desired target set while remaining inside the safe set until the first target hitting time. For deterministic systems, we develop three control guidance-barrier functions, namely exponential, asymptotic, and lax, to synthesize reach-avoid control inputs. These functions provide increasingly weak conditions, gradually expanding the space of admissible control inputs. The exponential and asymptotic control guidance-barrier functions are an extension of those proposed in [19] for inner-approximating reach-avoid sets. However, we observe that these two functions have a common condition that enforces invariance of every positive sublevel set until the system enters the target set, thus limiting the space of feasible controllers. To address this limitation, we utilize the boundary of the safe set to develop a lax control guidance-barrier function that relaxes this condition and obtains a weaker requirement, expanding the space of feasible reach-avoid controllers. Furthermore, we extend two control guidance-barrier functions to address stochastic systems. These functions facilitate the synthesis of a minimal reach-avoid controller, ensuring that the system eventually reaches a target set while remaining within a safe set until the first target hitting time, with a certain probability. One of the extended functions, known as the asymptotic control guidance-barrier function, is built upon the guidance-barrier function presented in [18], which was proposed for inner-approximating reach-avoid sets. The other function, referred to as the exponential control guidance-barrier function, is a novel contribution that generates convex constraints to enhance the efficient synthesis of reach-avoid controllers. Finally, we present several numerical examples to demonstrate the effectiveness of our proposed methods, using the semidefinite programming tool. Numerical results demonstrate that the proposed functions can work together in a complementary manner to improve the effectiveness of practical applications.

Our contributions are summarized as follows.

1) This article presents a unified framework for synthesizing reach-avoid controllers for both deterministic systems modeled by ordinary differential equations and stochastic systems modeled by stochastic differential equations in an optimal sense.
2) The framework proposes three control guidance-barrier functions—exponential, asymptotic, and lax—for constructing sufficient conditions to synthesize reach-avoid controllers for deterministic systems. These conditions gradually become less restrictive, allowing for an increasingly large set of controllers

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compatible with reach-avoid specifications. This provides more chances for successfully synthesizing a reach-avoid controller or improving an already obtained one. In addition, two of these functions are extended to stochastic systems.

The following notations will be used throughout the rest of this article: $\mathbb{R}^n$ denotes the set of n-dimensional real vectors, the closure of a set $\mathcal{X}$ is denoted by $\overline{\mathcal{X}}$, the complement by $\mathcal{X}^c$ and the boundary by $\partial\mathcal{X}$, $\|\boldsymbol{x}\|$ denotes the 2-norm, i.e., $\|\boldsymbol{x}\| := \sqrt{\sum_{i=1}^n x_i^2}$, where $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$, vectors are denoted by boldface letters. For a vector function $\boldsymbol{g}(\boldsymbol{x}) : \mathcal{C} \to \mathbb{R}^m$, $\|\boldsymbol{g}(\boldsymbol{x})\| = \sqrt{\int_{\boldsymbol{x}\in\mathcal{C}} \boldsymbol{g}^\top(\boldsymbol{x})\boldsymbol{g}(\boldsymbol{x})d\boldsymbol{x}}$.

Besides, due to space constraints, we have excluded the proofs in this article. However, interested readers can find them in the extended version, which is available at.[1] The proofs are actually based on those in [19] and [18].

II. Reach-Avoid Controllers Synthesis for Deterministic Systems

In this section, we focus our attention on solving the reach-avoid controllers synthesis problem for deterministic systems modeled by ordinary differential equations.

A. Preliminaries

In this section, we give an introduction on the system and reach-avoid controllers synthesis problem of interest.

Consider an affine control system,

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} \tag{1}$$

with $\boldsymbol{f}$ and $\boldsymbol{g}$ locally Lipschitz, $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{u} \in \mathcal{U} \subseteq \mathbb{R}^m$ is the set of admissible control inputs.

Let $\boldsymbol{u}(\boldsymbol{x}) : \mathbb{R}^n \to \mathcal{U}$ be a feedback controller such that the closed-loop system (1) is locally Lipschitz. Then, for any initial condition $\boldsymbol{x}_0 := \boldsymbol{x}(0) \in \mathbb{R}^n$, there exists a maximum time interval $I(\boldsymbol{x}_0, \boldsymbol{u}) = [0, \tau_{\max})$ such that $\boldsymbol{\phi}_{\boldsymbol{x}_0}(\cdot) : I(\boldsymbol{x}_0, \boldsymbol{u}) \to \mathbb{R}^n$ is the unique solution to system (1), where $\boldsymbol{\phi}_{\boldsymbol{x}_0}(0) = \boldsymbol{x}_0$ and $\tau_{\max}$ is the explosion time with $\lim_{t\to\tau_{\max}} \boldsymbol{\phi}_{\boldsymbol{x}_0}(t) = \infty$.

Given a bounded open safe set $\mathcal{C}$ and a target set $\mathcal{X}_r$, the reach-avoid property with respect to them and associated reach-avoid controllers are formulated in Definition 1.

Definition 1 (Reach-avoid Controllers): Given a feedback controller $\boldsymbol{u}(\cdot) : \mathbb{R}^n \to \mathcal{U}$, the reach-avoid property with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ is satisfied if, starting from any initial state in $\mathcal{C}$, system (1) with this controller will enter the target set $\mathcal{X}_r$ eventually while staying inside $\mathcal{C}$ before the first target hitting time. If $\boldsymbol{u}(\cdot) : \mathbb{R}^n \to \mathcal{U}$ is locally Lipschitz continuous, it is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$.

Then, we formulate our reach-avoid controllers synthesis problem.

Problem 1 (Reach-avoid Controllers Synthesis): Given a feedback control $\boldsymbol{u} = \boldsymbol{k}(\boldsymbol{x})$ for the control system (1), which may not be a reach-avoid controller, we wish to synthesize a reach-avoid controller in a minimally invasive fashion, i.e., modify the existing controller $\boldsymbol{k}(\boldsymbol{x})$ in a minimal way so as to guarantee satisfaction of the reach-avoid specification via solving the following optimization:

$$\begin{aligned} &\boldsymbol{u}(\boldsymbol{x}) = \arg\min \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| \\ &\text{s.t. } \boldsymbol{u}(\cdot) : \mathcal{C} \to \mathcal{U} \text{is a reach-avoid controller.} \end{aligned} \tag{2}$$

Besides, we impose the following assumptions on the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$.

Assumption 1: The safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ satisfy the following conditions.

1) $\mathcal{C} = \{\, \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) > 0 \,\}$.
2) $\overline{\mathcal{C}} = \{\, \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) \geq 0 \,\}$.
3) $\partial\mathcal{C} = \{\, \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) = 0 \,\}$, where $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is a continuously differentiable function, and
   1) $\mathcal{C}$ has no isolated point;
   2) $\mathcal{C} \cap \mathcal{X}_r$ is not empty and has no isolated point.

Although this work studies the reach-avoid controllers synthesis problem, the proposed methods can also be directly extended to the synthesis of controllers enforcing safety and stability, as in [3], with the assumption that the target set $\mathcal{X}_r$ is a region of attraction.

Remark 1: In this article, we focus our attention on the assumption that the safe set $\mathcal{C}$ is open. Some of our methods proposed in this article also work with a compact safe set $\mathcal{C}$. We will remark this extension in an appropriate place in the sequel. ■

B. Reach-Avoid Controllers Synthesis

In this section, we elucidate our methods on synthesizing reach-avoid controllers by solving optimization (2). The methods are built upon three control (i.e., exponential, asymptotic, and lax) guidance-barrier functions.

1) Exponential Control Guidance-Barrier Functions: In this subsection, we introduce exponential control guidance-barrier functions, which are derived from the exponential guidance-barrier function in [19].

Definition 2: Given the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ satisfying Assumption 1, $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is an exponential control guidance-barrier function if there exists $\lambda > 0$ such that

$$\sup_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U}} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq \lambda h(\boldsymbol{x}) \ \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$$

where $\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) = \bigtriangledown_{\boldsymbol{x}} h(\boldsymbol{x}) \cdot \boldsymbol{f}(\boldsymbol{x}) + \bigtriangledown_{\boldsymbol{x}} h(\boldsymbol{x}) \cdot \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}(\boldsymbol{x})$. ■

The exponential control guidance-barrier function will facilitate the construction of constraints for synthesizing reach-avoid controllers. The constraints are formulated in Theorem 1.

Theorem 1: Given the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ satisfying Assumption 1, if the function $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is an exponential control guidance-barrier function, then any Lipschitz continuous controller $\boldsymbol{u}(\cdot) : \mathcal{C} \to \mathcal{U}$ such that $\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{K}_e(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, where

$$\mathcal{K}_e(\boldsymbol{x}) = \{\, \boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U} \mid \text{constraint(3) holds} \}$$

with

$$\begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) - \lambda h(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \lambda > 0. \end{cases} \tag{3}$$

Remark 2: It is observed that the smaller λ is, the weaker constraint (3) is. However, λ cannot be zero. It is also observed that an exponential control guidance-barrier function complements a zeroing control barrier function satisfying

$$\begin{cases} \sup_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U}} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq \lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{\mathcal{C}} \\ \lambda < 0. \end{cases}.$$

A zeroing control barrier function simplifies controller design by focusing solely on safety enforcement. Alternatively, a single control exponential guidance-barrier function can achieve both safety and stability goals by functioning as both a CBF and a control Lyapunov function, as long as the region of stabilization $\mathcal{X}_r$ is defined. Although [3] combined control barrier functions with control Lyapunov functions to achieve safety and stability, this alternative approach offers a more streamlined solution. ■

The set $\mathcal{K}_e$ is convex, thus a direct computation of a locally Lipschitz continuous feedback controller $\boldsymbol{u}(\boldsymbol{x})$ satisfying constraint (3) using convex optimization is possible. Based on the exponential guidance-barrier function $h(\boldsymbol{x})$ and Theorem 1, Problem 1 could be addressed via solving the following program:

$$\begin{aligned} &(\boldsymbol{u}_0(\boldsymbol{x}), \delta) = \arg\min_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U},\lambda,\delta} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| + c\delta \\ &\text{s.t.} \begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) - \lambda h(\boldsymbol{x}) \geq -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \delta \geq 0 \\ \lambda \geq \xi_0 \end{cases} \end{aligned} \tag{4}$$

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where $\boldsymbol{u}(\boldsymbol{x}) : \overline{\mathcal{C}} \to \mathcal{U}$ is a locally Lipschitz parameterized function with some unknown parameters, ξ_0 is a user-defined positive value which is to enforce the strict positivity of λ, and c is a large constant that penalizes safety/reachability violations. As commented in Remark 2, the smaller ξ_0 is, the weaker the constraint in (4) is. However, ξ_0 cannot be zero. Thus, in numerical computations, we do not recommend the use of too small ξ_0 due to numerical errors. The optimization is not sensitive to the c value as long as it is very large (e.g., 10^{12}), such that the constraint violations are heavily penalized. When $\delta > 0$, it provides graceful degradation when the reach-avoid objective cannot be enforced.

Proposition 1: Suppose $(\boldsymbol{u}_0(\boldsymbol{x}), \delta)$ is obtained by solving optimization (4), and $\boldsymbol{u}_0(\boldsymbol{x})$ is locally Lipschitz continuous,

1) If $\delta = 0$, $\boldsymbol{u}_0(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$.
2) If $\delta > 0$ and the set $\mathcal{D} = \{ \boldsymbol{x} \mid h(\boldsymbol{x}) > \frac{\delta}{\lambda} \}$ has nonempty intersection with the target set $\mathcal{X}_r$, $\boldsymbol{u}_0(\boldsymbol{x})$ is the reach-avoid controller with respect to the set $\mathcal{D}$ and target set $\mathcal{X}_r$.

If $\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{K}_e(\boldsymbol{x})$, it will drive system (1) to enter the target set $\mathcal{X}_r$ at an exponential rate of λ. Besides, it requires $h(\boldsymbol{\phi}_{\boldsymbol{x}_0}(t))$ to be strictly monotonically increasing with respect to t before the trajectory $\boldsymbol{\phi}_{\boldsymbol{x}_0}(t)$ enters the target set $\mathcal{X}_r$. Thus, if there exists $\boldsymbol{y} \in \mathcal{C} \setminus \mathcal{X}_r$ such that $h(\boldsymbol{y}) \geq \sup_{\boldsymbol{x} \in \mathcal{X}_r} h(\boldsymbol{x})$, we cannot obtain a reach-avoid controller with respect to the safe set $\mathcal{C}$ and the target set $\mathcal{X}_r$ via solving (4). Therefore, solving (4) may either obtain a pessimistic reach-avoid controller or lead to $\delta > 0$. In the sequel, we will obtain two new optimizations with weaker constraints for synthesizing reach-avoid controllers when $\lambda = 0$ and $\lambda < 0$.

Remark 3: When the safe set $\mathcal{C}$ is compact, i.e., $\mathcal{C} = \{ \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) \geq 0 \}$, a slight modification to constraint (3) is to replace the "greater than or equal to sign" with the "greater sign" in the first condition $\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) - \lambda h(\boldsymbol{x}) \geq 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$, i.e.,

$$\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) - \lambda h(\boldsymbol{x}) > 0 \ \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$$

where $\lambda > 0$. This modification is to enforce the reachability objective for states on the boundary $\partial\mathcal{C}$. ■

2) Asymptotic Control Guidance-Barrier Functions: In this subsection, we introduce asymptotic control guidance-barrier functions for the synthesis of reach-avoid controllers.

Definition 3: Given the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ satisfying Assumption 1, then $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is called an asymptotic control guidance-barrier function, if there exists a continuously differentiable function $w(\cdot) : \mathbb{R}^n \to \mathbb{R}$ such that

$$\sup_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}} \min \{ \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}), \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) \} \geq 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}. \quad ■$$

The asymptotic control guidance-barrier function also facilitates the construction of constraints for synthesizing reach-avoid controllers. The constraints are formulated in Theorem 2.

Theorem 2: Given the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ satisfying Assumption 1, if the function $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is an asymptotic control guidance-barrier function, then any Lipschitz continuous controller $\boldsymbol{u}(\cdot) : \mathcal{C} \to \mathcal{U}$ such that $\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{K}_a(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, where

$$\mathcal{K}_a(\boldsymbol{x}) = \{ \boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U} \mid \text{constraint (5) holds} \}$$

with

$$\begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}. \end{cases} \tag{5}$$

According to Theorem 2, Problem 1 could be addressed via solving the following optimization:

$$\boldsymbol{u}(\boldsymbol{x}) = \arg \min_{\boldsymbol{u} \in \mathbb{R}^m} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| + c\delta$$

$$\text{s.t.} \quad \begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \delta \geq 0 \end{cases} \tag{6}$$

Algorithm 1: An iterative procedure for solving optimization (6) when a reach-avoid controller is returned via solving optimization (4).

Let $\boldsymbol{u}_0(\cdot) : \mathbb{R}^n \to \mathcal{U}$ be computed via solving optimization (4), and $\epsilon' > 0$ be a specified threshold.
$\xi_0 := \|\boldsymbol{u}_0(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$;
$i := 0$;
while TRUE **do**
compute $w_i^*(\boldsymbol{x})$ such that $\mathcal{L}_{w_i,\boldsymbol{u}_i}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq 0 \ \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$;
solve the following program to obtain $\boldsymbol{u}_{i+1}(\boldsymbol{x})$:

$$\min_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$$

$$\text{s.t.} \begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \mathcal{L}_{w_i^*,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \end{cases};$$

if $\xi_{i+1} - \xi_i \leq -\epsilon'$, where $\xi_{i+1} = \|\boldsymbol{u}_{i+1}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$ **then**
$i := i + 1$;
else
return $\boldsymbol{u}_{i+1}(\boldsymbol{x})$ and terminate;
end if
end while

where $\boldsymbol{u}(\boldsymbol{x}) : \overline{\mathcal{C}} \to \mathcal{U}$ is a locally Lipschitz parameterized function with some unknown paramteres, and c is a large constant that penalizes reachability violations.

Via comparing constraints (5) and (3), it is easy to observe that constraint (5) is more expressive, and thus, admits more feasible feedback controllers, since constraint (5) will be reduced to (3) when $w(\boldsymbol{x}) = \frac{h(\boldsymbol{x})}{\lambda}$. Consequently, it is more likely to either obtain a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, or improve the obtained reach-avoid controller when using constraint (5) for computations, and thus, lead to smaller $\|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$. Besides, compared to exponential control guidance-barrier functions, a reach-avoid controller generated by an asymptotic control guidance-barrier function requires $h(\boldsymbol{\phi}_{\boldsymbol{x}_0}(t))$ to be monotonically nondecreasing with respect to t rather than strictly monotonically increasing before the trajectory $\boldsymbol{\phi}_{\boldsymbol{x}_0}(t)$ enters the target set $\mathcal{X}_r$, thereby diversifying the application scenarios. However, the synthesis of a reach-avoid controller $\boldsymbol{u}(\boldsymbol{x})$ via solving constraint (5) is a nonconvex problem due to the presence of the term $\bigtriangledown_{\boldsymbol{x}} w(\boldsymbol{x}) \cdot g(\boldsymbol{x})\boldsymbol{u}$. We can adopt an iterative algorithm to solve optimization (6) by decomposing it into two convex subproblems based on two cases, which correspond to whether a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ is obtained via solving optimization (6) or not.

The iterative algorithm for solving optimization (6) to obtain a suboptimal controller is elucidated in the following.

Case 1: A Reach-Avoid Controller is Returned Via Solving Optimization (4), that is, $\delta = 0$ in (4). According to the comparison between constraints (5) and (3) above, a reach-avoid controller $\boldsymbol{u}_0(\boldsymbol{x})$ from optimization (4) is also a solution to optimization (6). Consequently, we can adopt the iterative algorithm in Algorithm 1 to solve optimization (6) for improving the controller $\boldsymbol{u}_0(\boldsymbol{x})$.

Case 2: A Reach-Avoid Controller is not Returned Via Solving Optimization (4), that is, $\delta > 0$ in (4). When a reach-avoid controller $\boldsymbol{u}(\boldsymbol{x})$ is not computed via solving optimization (4), one reason leading to this failure is due to the strictness of constraint (3) and the other one is the potential conflict between reachability and safety. At all events, in safety-critical systems the safety has the highest priority.

One of advantages of constraint (5) over (3) lies in its ability of characterizing safety and reachability separately. The condition

$$\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 \ \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \tag{7}$$

ensures safety, i.e., trajectories starting from $\mathcal{C} \setminus \mathcal{X}_r$ will either stay inside the set $\mathcal{C} \setminus \mathcal{X}_r$ for all the time or enter the target set $\mathcal{X}_r$ eventually

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while staying inside the safe set $\mathcal{C}$ before the first target hitting time. We term the Lipschitz controller $\boldsymbol{u}(\boldsymbol{x})$ satisfying constraint (7) a safe controller. On the other hand, the condition

$$\mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq 0 \; \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \tag{8}$$

ensures reachability, i.e., trajectories starting from $\mathcal{C} \setminus \mathcal{X}_r$ will eventually enter the target set $\mathcal{X}_r$. It is worth remarking that a function $w(\boldsymbol{x})$ satisfying constraint (8) is not a Lyapunov function, as in [3], since the function $w(\boldsymbol{x})$ itself is not required to be sign definite.

Constraint (7) is convex, i.e., the set of controllers satisfying constraint (7) is convex. Therefore, we first synthesize a safe controller to ensure safety of system (1) via solving the following optimization:

$$\boldsymbol{u}^*(\boldsymbol{x}) = \arg \min_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$$

$$\text{s.t.} \quad \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 \; \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}. \tag{9}$$

After computing a safe controller $\boldsymbol{u}^*(\boldsymbol{x})$, we then solve the following optimization:

$$(w^*(\boldsymbol{x}), \delta^*) = \arg \min_{w(\boldsymbol{x}), \delta} \delta$$

$$\text{s.t.} \begin{cases} \mathcal{L}_{w,\boldsymbol{u}^*}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \delta \geq 0. \end{cases} \tag{10}$$

If $\delta^* = 0$, the controller $\boldsymbol{u}^*(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$. Otherwise, the controller $\boldsymbol{u}^*(\boldsymbol{x})$ degrades the reachability performance.

Proposition 2: Suppose $\delta^* \in [0, \infty)$ is obtained by solving optimization (10). If the set $\mathcal{D} = \{ \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) > \delta^* \}$ has a nonempty intersection with the target set $\mathcal{X}_r$, the controller $\boldsymbol{u}^*(\boldsymbol{x})$ is a reach-avoid controller with respect to $\mathcal{D}$ and $\mathcal{X}_r$.

If $\delta^* > 0$, then we proceed to obtain a new Lipschitz controller $\boldsymbol{u}^{**}(\boldsymbol{x})$ and a smaller δ^{**} via solving optimization (11) with $w^*(\boldsymbol{x})$

$$(\boldsymbol{u}^{**}(\boldsymbol{x}), \delta^{**}) = \arg \min_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}, \delta} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| + c\|\delta\|$$

$$\text{s.t.} \quad \begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \mathcal{L}_{w^*,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \delta \geq 0 \end{cases} \tag{11}$$

where c is a large constant that penalizes reachability violations, similar to the one in optimization (4).

Repeating the above procedure leads to an iterative algorithm for solving optimization (6), which is summarized in Algorithm 2. When $\delta = 0$, a reach-avoid controller $\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{K}_a(\boldsymbol{x})$ is generated successfully.

Although constraint (5) is weaker than (3), it is still strict in practical applications. The condition $\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$ not only ensures invariance of the set $\overline{\mathcal{C} \setminus \mathcal{X}_r}$ but also enforces invariance of every positive sublevel set of the asymptotic control guidance-barrier function (if trajectories do not enter the target set $\mathcal{X}_r$). This is overly conservative. Thus, solving optimization (6) may lead to either a failure in synthesizing a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, correpsponding to $\delta > 0$, or a pessimistic feedback controller, i.e., $\|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$ is large.

Remark 4: Like zeroing control barrier functions in [3], a relaxation of the first condition, i.e., $\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$, in constraint (5) is

$$\begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \lambda > 0. \end{cases}$$

In such circumstances, we have to intensify the second condition, i.e., $\mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) \geq 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$, in constraint (5) using $\mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) - h(\boldsymbol{x}) > 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$. Further, since $h(\boldsymbol{x}) \geq 0$ for $\boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$, the above constraints can be weakened by

$$\begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) > 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \lambda > 0. \end{cases} \tag{12}$$

Algorithm 2: An iterative procedure for solving optimization (6) when a reach-avoid controller is not returned via solving optimization (4).

Let $\epsilon', \epsilon^* > 0$ be specified thresholds.
$i := 0$;
solve optimization (9) to obtain a safe controller $\boldsymbol{u}_i(\boldsymbol{x})$;
while TRUE **do**
compute $(w_i(\boldsymbol{x}), \delta_i)$ via solving optimization (10) with $\boldsymbol{u}_i(\boldsymbol{x})$;
compute $(\boldsymbol{u}_{i+1}(\boldsymbol{x}), \delta_{i+1})$ via solving (11) with $w_i(\boldsymbol{x})$;
if $\delta_i = \delta_{i+1} = 0$ **then**
if $\xi_{i+1} - \xi_i \leq -\epsilon'$, where $\xi_{i+1} = \|\boldsymbol{u}_{i+1}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$ **then**
$i := i + 1$;
else
return $\boldsymbol{u}_{i+1}(\boldsymbol{x})$ and terminate;
end if
else
if $\delta_{i+1} - \delta_i \leq -\epsilon^*$ **then**
$i := i + 1$;
else
return $\boldsymbol{u}_{i+1}(\boldsymbol{x})$ and terminate;
end if
end if
end while

Constraint (12) is nearly identical with the ones in control Lyapunov-Barrier functions in [3]. One difference is that $w(\boldsymbol{x})$ is unknown in constraint (12). However, if we replace the constraint in (9) with

$$\begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \lambda > 0 \end{cases}$$

to obtain a safe controller $\boldsymbol{u}^*(\boldsymbol{x})$, we cannot obtain a conclusion as in Proposition 2 via solving optimization (10) with the constraints $\mathcal{L}_{w,\boldsymbol{u}^*}(\boldsymbol{x}) > -\delta, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$ and $\delta \geq 0$ when $\delta^* > 0$. ■

3) Lax Control Guidance-Barrier Functions: In this subsection, we will further relax constraints in asymptotic guidance-barrier functions and obtain less restrictive ones, expanding the space of controllers compatible with reach-avoid specifications. These new constraints are based on a subset of the safe set $\mathcal{C}$, which we call the tightened set $\mathcal{D}$.

The tightened set $\mathcal{D}$ can be any subset of the set $\mathcal{C}$ satisfying the following conditions.

1) $\mathcal{D} \subseteq \mathcal{C}$.
2) $\partial\mathcal{D} \cap \partial\mathcal{C} = \emptyset$.
3) $\mathcal{D} \cap \mathcal{X}_r \neq \emptyset$.

In this article, we take $\mathcal{D} = \{ \boldsymbol{x} \in \mathbb{R}^n \mid h'(\boldsymbol{x}) \geq 0 \}$, where $h'(\boldsymbol{x}) = h(\boldsymbol{x}) - \epsilon_0$ with $\epsilon_0 > 0$. It is observed that ϵ_0 can be an arbitrary positive value, and $\lim_{\epsilon_0 \to 0^+} \mathcal{D} = \mathcal{C}$.

Using the set $\mathcal{D}$, we define a lax control guidance-barrier function.

Definition 4: Let $\mathcal{C} \subseteq \mathbb{R}^n$ and $\mathcal{X}_r$ satisfy Assumption 1 and $\mathcal{D}$ be the tightened set, then $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is called a lax control guidance-barrier function, if there exist a continuously differentiable function $w(\cdot) : \mathbb{R}^n \to \mathbb{R}$ and $\lambda \geq 0$ such that

$$\begin{cases} \sup_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{(\mathcal{C} \setminus \mathcal{D}) \setminus \mathcal{X}_r} \\ \sup_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}} \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) > 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}. \end{cases} \tag{13}$$

Theorem 3: Let $\mathcal{C} \subseteq \mathbb{R}^n$ and $\mathcal{X}_r$ satisfy Assumption 1 and $\mathcal{D}$ be a tightened set defined above, if $h(\boldsymbol{x})$ is a lax control guidance-barrier function, then any Lipschitz continuous controller $\boldsymbol{u}(\cdot) : \mathcal{C} \to \mathcal{U}$ satisfying $\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{K}_l(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, where

$$\mathcal{K}_l(\boldsymbol{x}) = \{ \boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U} \mid \text{constraint (14) holds} \}$$

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where

$$\begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{(\mathcal{C} \setminus \mathcal{D}) \setminus \mathcal{X}_r} \\ \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) > 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \lambda \geq 0. \end{cases} \tag{14}$$

Remark 5: Constraint (14) also applies to the case that the safe set $\mathcal{C}$ is compact. ■

In constraint (14), when $\lambda = 0$, the condition

$$\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) \, \forall \boldsymbol{x} \in \overline{(\mathcal{C} \setminus \mathcal{D}) \setminus \mathcal{X}_r}$$

ensures invariance of the set $\overline{\mathcal{D}' \setminus \mathcal{X}_r}$ until system (1) enters the target set $\mathcal{X}_r$, where $\mathcal{D}' = \{ \boldsymbol{x} \in \mathcal{C} \mid h(\boldsymbol{x}) \leq \epsilon_1 \}$ with $\epsilon_1 \leq \epsilon_0$. It does not ensure invariance of the set $\{ \boldsymbol{x} \in \mathcal{C} \mid h(\boldsymbol{x}) \geq \epsilon_1 \}$ with $\epsilon_1 > \epsilon_0$, and thus, it is weaker than the condition $\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$ in constraint (5). However, when $\lambda > 0$, we can just ensure invariance of the set $\mathcal{C} \setminus \mathcal{X}_r$ until system (1) enters the target set $\mathcal{X}_r$. However, since $h(\boldsymbol{x}) \geq 0$ for $\boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$, we cannot conclude that if $\mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) \geq h(\boldsymbol{x}) \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$ holds, $\mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) > 0 \, \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}$ holds. It is worth noting that constraint (14) is weaker than (12). Moreover, unlike exponential and asymptotic control guidance-barrier functions, a reach-avoid controller generated by a lax one does not require $h(\boldsymbol{\phi}_{\boldsymbol{x}_0}(t))$ to be monotonically increasing with respect to t before the trajectory $\boldsymbol{\phi}_{\boldsymbol{x}_0}(t)$ enters the target set $\mathcal{X}_r$. This diversifies the application scenarios even further.

Example 1: Consider a simple system

$$\begin{cases} \dot{x} = -x + 0.3 \\ \dot{y} = -y + u_1 \end{cases} \tag{15}$$

where $\mathcal{C} = \{ (x,y)^\top \mid 1 - x^2 - y^2 > 0 \}$, $\mathcal{X}_r = \{ (x,y)^\top \mid (x-0.3)^2 + y^2 - 0.01 < 0 \}$, and $\mathcal{U} = \{ u_1 \mid -0.1 \leq u_1 \leq 0.1 \}$.

Since $h(0,0) = 1$ and $h(x,y) < 1$ for $(x,y)^\top \in \mathcal{X}_r$, we cannot obtain a reach-avoid controller with respect to the safe set $\mathcal{C}$ and the target set $\mathcal{X}_r$ with exponential and asymptotic guidance-barrier functions. However, we can obtain a reach-avoid controller $u_1 = 0$ with lax guidance-barrier functions. $h(x,y)$ is a lax guidance-barrier function satisfying (14) with $\lambda = 2$, $w(x,y) = -(x-0.3)^2 - y^2$, $\mathcal{D} = \{ (x,y)^\top \mid 0.99 - x^2 - y^2 \geq 0 \}$, and $u_1 = 0$. ■

According to Proposition 3, Problem 1 can be transformed into the following optimization:

$$\boldsymbol{u}(\boldsymbol{x}) = \arg \min_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}, \delta, \lambda} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| + c\delta$$

$$\text{s.t.} \quad \begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{(\mathcal{C} \setminus \mathcal{D}) \setminus \mathcal{X}_r} \\ \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) > -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \delta \geq 0 \\ \lambda \geq 0 \end{cases} \tag{16}$$

where $\boldsymbol{u}(\boldsymbol{x}) : \overline{\mathcal{C}} \to \mathcal{U}$ is a locally Lipschitz parameterized function with some unknown paramteres, and c is a large constant that penalizes reachability violations.

Similar to optimization (6), optimization (16) is also nonconvex. Thus, we also decompose it into two convex subproblems and solve it using an iterative algorithm, which is described in Algorithm 3.

Remark 6: Suppose that $\delta_i (i = 1, \ldots)$ is computed by solving optimization (18) or (19). If $\delta_i = 0$, then the controller $\boldsymbol{u}_i(\boldsymbol{x})$ satisfies Theorem 3. Otherwise, we can neither ensure that the controller $\boldsymbol{u}_i(\boldsymbol{x})$ will drive system (1) starting from $\mathcal{C}$ to enter the target set $\mathcal{X}_r$ eventually nor ensure that $\boldsymbol{u}_i(\boldsymbol{x})$ will drive system (1) starting from $\{ \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) > \delta_i \}$ to enter the target set $\mathcal{X}_r$ eventually.

Remark 7: We notice that the constraint (14) weakens as the value of ϵ_0 used to define $\mathcal{D}$ decreases, but it cannot be set to zero. While it may be tempting to use a very small value for ϵ_0 in numerical computations to strengthen the constraint, this approach should be avoided due to potential numerical errors.

Algorithm 3: An iterative procedure for solving optimization (16).

Let $\epsilon', \epsilon^* > 0$ be specified thresholds.
$i := 0$;
if a Lipschitz controller $\boldsymbol{u}(\boldsymbol{x})$ is computed via solving optimization (6) **then**
$\boldsymbol{u}_0(\boldsymbol{x}) := \boldsymbol{u}(\boldsymbol{x})$;
else
compute a Lipschitz safe controller $\boldsymbol{u}_0(\boldsymbol{x})$ satisfying

$$\min_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$$

$$\text{s.t.} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 \; \forall \boldsymbol{x} \in \overline{(\mathcal{C} \setminus \mathcal{D}) \setminus \mathcal{X}_r}. \tag{17}$$

end if
while TRUE **do**
compute $(w_i(\boldsymbol{x}), \delta_i)$ satisfying

$$\min_{w(\boldsymbol{x}), \delta} \delta$$

$$\text{s.t.} \begin{cases} \mathcal{L}_{w,\boldsymbol{u}_i}(\boldsymbol{x}) > -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \delta \geq 0 \end{cases} \tag{18}$$

solve the following program to obtain $(\boldsymbol{u}_{i+1}(\boldsymbol{x}), \delta_{i+1})$:

$$\min_{\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U}, \delta, \lambda} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| + c\delta$$

$$\text{s.t.} \begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq -\lambda h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{(\mathcal{C} \setminus \mathcal{D}) \setminus \mathcal{X}_r} \\ \mathcal{L}_{w_i^*,\boldsymbol{u}}(\boldsymbol{x}) > -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \delta \geq 0 \\ \lambda \geq 0 \end{cases} \tag{19}$$

if $\delta_i = \delta_{i+1} = 0$ **then**
if $\xi_{i+1} - \xi_i \leq -\epsilon'$, where $\xi_{i+1} = \|\boldsymbol{u}_{i+1}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$ **then**
$i := i + 1$;
else
return $\boldsymbol{u}_{i+1}(\boldsymbol{x})$ and terminate;
end if
else
if $\delta_{i+1} - \delta_i \leq -\epsilon^*$ **then**
$i := i + 1$;
else
return $\boldsymbol{u}_{i+1}(\boldsymbol{x})$ and terminate;
end if
end if
end while

III. Reach-Avoid Controllers Synthesis for Stochastic Systems

This section focuses on synthesizing reach-avoid controllers for stochastic systems modeled by stochastic differential equations.

A. Preliminaries

In this subsection, we introduce stochastic systems and reach-avoid controllers synthesis problems of interest.

Consider an affine stochastic control system,

$$d\boldsymbol{x}(t,\boldsymbol{w}) = (\boldsymbol{f}(\boldsymbol{x}(t,\boldsymbol{w})) + \boldsymbol{g}(\boldsymbol{x}(t,\boldsymbol{w}))\boldsymbol{u}(\boldsymbol{x}(t)))dt + \boldsymbol{\sigma}(\boldsymbol{x}(t,\boldsymbol{w}))d\boldsymbol{W}(t,\boldsymbol{w}) \tag{20}$$

where $\boldsymbol{f}(\cdot) : \mathbb{R}^n \to \mathbb{R}^n$, $\boldsymbol{g}(\cdot) : \mathbb{R}^n \to \mathbb{R}^{n \times m}$, and $\boldsymbol{\sigma}(\cdot) : \mathbb{R}^n \to \mathbb{R}^{n \times k}$ are locally Lipschitz continuous functions. $\boldsymbol{u}(\cdot) : \mathbb{R}^n \to \mathcal{U}$ with $\mathcal{U} \subset \mathbb{R}^m$ is the admissible input. $\boldsymbol{W}(t,\boldsymbol{w}) : \mathbb{R} \times \Omega \to \mathbb{R}^k$ is a k-dimensional Wiener process (standard Brownian motion), and Ω,

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equipped with the probability measure $\mathbb{P}$, is the sample space $\boldsymbol{w}$ belongs to. The expectation with respect to $\mathbb{P}$ is denoted by $\mathbb{E}[\cdot]$.

Assume that each component of $\boldsymbol{f}(\boldsymbol{x})$, $\boldsymbol{g}(\boldsymbol{x})$, and $\boldsymbol{\sigma}(\boldsymbol{x})$ is locally Lipschitz continuous. Given a locally Lipschitz controller $\boldsymbol{u}(\boldsymbol{x})$, then for an initial state $\boldsymbol{x}_0 \in \mathbb{R}^n$, a stochastic differential equation (20) has a unique (maximal local) strong solution over some time interval $[0, T^{\boldsymbol{x}_0}(\boldsymbol{w}))$ for $\boldsymbol{w} \in \Omega$, where $T^{\boldsymbol{x}_0}(\boldsymbol{w})$ is a positive real value or infinity. We denote it as $\boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(\cdot) : [0, T^{\boldsymbol{x}_0}(\boldsymbol{w})) \to \mathbb{R}^n$, which satisfies the stochastic integral equation

$$\boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(t) = \boldsymbol{x}_0 + \int_0^t (\boldsymbol{f}(\boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(\tau)) + \boldsymbol{g}(\boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(\tau))\boldsymbol{u}(\boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(\tau)))d\tau + \int_0^t \boldsymbol{\sigma}(\boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(\tau))d\boldsymbol{W}(\tau, \boldsymbol{w}).$$

The infinitesimal generator underlying system (20) is presented in Definition 5.

Definition 5: Given system (20) with a locally Lipschitz controller $\boldsymbol{u}(\boldsymbol{x})$, the infinitesimal generator of a twice continuously differentiable function $v(\boldsymbol{x})$ is defined by

$$\begin{aligned}\mathcal{L}_{v,\boldsymbol{u}}(\boldsymbol{x}_0) &= \lim_{t\to 0} \frac{\mathbb{E}[v(\boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(t))] - v(\boldsymbol{x}_0)}{t} \\ &= \left[\frac{\partial v}{\partial \boldsymbol{x}}(\boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}(\boldsymbol{x})) + \frac{1}{2}\mathbf{tr}\left(\boldsymbol{\sigma}(\boldsymbol{x})^\top \frac{\partial^2 v}{\partial \boldsymbol{x}^2}\boldsymbol{\sigma}(\boldsymbol{x})\right)\right]\Bigg|_{\boldsymbol{x}=\boldsymbol{x}_0}.\end{aligned}$$

Given a bounded and open safe set

$$\mathcal{C} = \{\, \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) > 0 \,\}$$

with $\partial\mathcal{C} = \{\, \boldsymbol{x} \in \mathbb{R}^n \mid h(\boldsymbol{x}) = 0 \,\}$, and a target set $\mathcal{X}_r$ satisfying $\mathcal{X}_r \subseteq \mathcal{C}$ and

$$h(\boldsymbol{x}) \leq 1 \,\forall \boldsymbol{x} \in \mathcal{X}_r$$

a reach-avoid controller is formulated in Definition 6.

Definition 6 (Reach-Avoid Controllers): Given a locally Lipschitz continuous feedback controller $\boldsymbol{u}(\cdot) : \mathbb{R}^n \to \mathcal{U}$, the reach-avoid property with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ is satisfied if, starting from any initial state $\boldsymbol{x}_0$ in $\mathcal{C}$, system (20) with the controller $\boldsymbol{u}(\cdot)$ will enter the target set $\mathcal{X}_r$ eventually while staying inside $\mathcal{C}$ before the first target hitting time, with a probability being larger than $h(\boldsymbol{x}_0)$, i.e.,

$$\mathbb{P}\left(\left\{\boldsymbol{w} \in \Omega \mid \begin{array}{l}\exists t \geq 0. \boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(t) \in \mathcal{X}_r \bigwedge \\ \forall \tau \in [0,t]. \boldsymbol{\phi}^{\boldsymbol{w}}_{\boldsymbol{x}_0}(\tau) \in \mathcal{C}\end{array}\right\}\right) \geq h(\boldsymbol{x}_0).$$

Correspondingly, $\boldsymbol{u}(\cdot) : \mathbb{R}^n \to \mathcal{U}$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$.

Similar to the case for deterministic systems in Section II, we wish to synthesize a reach-avoid controller in a minimally invasive fashion. The optimization problem is formulated as follows.

Problem 2: Suppose we are given a feedback controller $\boldsymbol{k}(\boldsymbol{x}) : \mathcal{C} \to \mathbb{R}^m$ for system (20), we wish to modify this controller in a minimal way so as to guarantee satisfaction of reach-avoid specifications, i.e., solve the following optimization:

$$\begin{aligned}&\boldsymbol{u}(\boldsymbol{x}) = \arg\min \|\boldsymbol{u} - \boldsymbol{k}(\boldsymbol{x})\| \\ &\text{s.t. } \boldsymbol{u}(\cdot) : \mathbb{R}^n \to \mathcal{U} \text{is a reach} - \text{avoid controller} \\ &\quad \text{with respect to the safe set } \mathcal{X} \text{and target set} \mathcal{X}_r.\end{aligned} \tag{21}$$

B. Reach-Avoid Controllers Synthesis

In this section, we extend our methods for deterministic systems in Section II-B1 and II-B2 to stochastic systems modeled by stochastic differential equations (20). The algorithms of solving the resulting optimizations are the same with the ones in Section II-B1 and II-B2 and therefore, are omitted here.

1) Exponential Control Guidance-Barrier Functions: In this subsection, we introduce the notion of exponential control guidance-barrier functions for synthesizing reach-avoid controllers.

Definition 7: Let $\mathcal{C}$ and $\mathcal{X}_r$ be the sets defined in Section III-A, then $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is called an exponential control guidance-barrier function, if there exists $\alpha \in (0, \infty)$ such that

$$\sup_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U}} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq \alpha h(\boldsymbol{x}) \;\forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}. \tag{22}$$

Given an exponential control guidance-barrier function, the sufficient condition for ensuring that a controller $\boldsymbol{u}(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$ is formulated in Theorem 4.

Theorem 4: Given the safe set $\mathcal{C}$, target set $\mathcal{X}_r$ defined in Section III-A, if $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is an exponential control guidance-barrier function, then any locally Lipschitz controller $\boldsymbol{u}(\cdot) : \overline{\mathcal{C}} \to \mathcal{U}$ satisfying $\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{K}_{s,e}(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, where

$$\mathcal{K}_{s,e}(\boldsymbol{x}) = \{\, \boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U} \mid \text{constraint (23) holds}\}$$

with

$$\begin{cases}\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq \alpha h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \alpha > 0.\end{cases} \tag{23}$$

Similarly, optimization (21) can be encoded into the following program:

$$\min_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U},\alpha,\delta} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| + c\delta$$

$$\text{s.t.} \begin{cases}\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) - \alpha h(\boldsymbol{x}) \geq -\delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ 0 \leq \delta < \alpha \\ \alpha \geq \xi_0\end{cases}, \tag{24}$$

where $\boldsymbol{u}(\boldsymbol{x}) : \overline{\mathcal{C}} \to \mathcal{U}$ is a locally Lipschitz parameterized function with some unknown parameters, ξ_0 is a user-defined positive value which is to enforce the strict positivity of α, and c is a large constant that penalizes safety/reachability violations. The set $\mathcal{K}_{s,e}(\boldsymbol{x})$ is convex, thus a direct computation of a feedback controller $\boldsymbol{u}(\boldsymbol{x})$ satisfying constraints in (24) using convex optimization is possible.

Proposition 3: Suppose $(\boldsymbol{u}_0(\boldsymbol{x}), \alpha, \delta)$ is obtained by solving optimization (24).

1) If $\delta = 0$, $\boldsymbol{u}_0(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$.
2) If $\delta \neq 0$, $\boldsymbol{u}_0(\boldsymbol{x})$ is a reach-avoid controller with respect to the set $\mathcal{D}$ and target set $\mathcal{X}_r$, where $\mathcal{D} = \{\, \boldsymbol{x} \mid h'(\boldsymbol{x}) > 0 \,\}$ with $h'(\boldsymbol{x}) = \frac{\alpha h(\boldsymbol{x}) - \delta}{\alpha - \delta}$.

2) Asymptotic Control Guidance-Barrier Functions: In this subsection, we introduce asymptotic control guidance-barrier functions for synthesizing reach-avoid controllers.

Definition 8: Given sets $\mathcal{C}$ and $\mathcal{X}_r$ defined in Section III-A, then $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is called an asymptotic control guidance-barrier function, if there exists a twice continuously differentiable function $w(\cdot) : \mathbb{R}^n \to \mathbb{R}$ such that

$$\begin{cases}\sup_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U}} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \sup_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U}} \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) \geq h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}.\end{cases} \tag{25}$$

The asymptotic control guidance-barrier function facilitates the construction of constraints for synthesizing reach-avoid controllers.

Theorem 5: Given sets $\mathcal{C}$ and $\mathcal{X}_r$ defined in Section III, if $h(\cdot) : \mathbb{R}^n \to \mathbb{R}$ is an asymptotic control guidance-barrier function, then any Lipschitz controller $\boldsymbol{u}(\cdot) : \mathcal{C} \to \mathcal{U}$ satisfying $\boldsymbol{u}(\boldsymbol{x}) \in \mathcal{K}_{s,a}(\boldsymbol{x})$ is a reach-avoid controller with respect to the safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, where

$$\mathcal{K}_{s,a}(\boldsymbol{x}) = \{\, \boldsymbol{u}(\boldsymbol{x}) \in \mathcal{U} \mid \text{constraint(26) holds}\}$$

with

$$\begin{cases}\mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) \geq h(\boldsymbol{x}) & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r}.\end{cases} \tag{26}$$

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TABLE I
PARAMETERS IN SOLVING OPTIMIZATION (4), (6), (16), (24), AND (27) (ϵ_0 IS USED TO DEFINE $\mathcal{D}$)

Optimization		
(4)/(24)	(6)/(27)	(16)
$\xi_0 = 10^{-3}$	$\epsilon' = 10^{-3}$, $\epsilon^* = 10^{-3}$	$\epsilon' = 10^{-3}$, $\epsilon^* = 10^{-3}$

TABLE II
$\|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$ FOR EXAMPLES 2–4

Example	Optimization		
	(4)	(6)	(16)
2	0.2329	0.089	3.829×10^{-4}
3	3.2002	3.1563	3.1563
4	1.7967	1.7912	1.7372

TABLE III
$\|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$ FOR EXAMPLE 5

Example	Optimization	
	(24)	(27)
5	93.6528	93.4427

According to Theorem 5, optimization (21) can be encoded into the following program:

$$\min_{\boldsymbol{u}(\boldsymbol{x})\in\mathcal{U}, w(\boldsymbol{x}), \delta} \|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\| + c\delta$$

$$\text{s.t.} \quad \begin{cases} \mathcal{L}_{h,\boldsymbol{u}}(\boldsymbol{x}) \geq 0 & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ \mathcal{L}_{w,\boldsymbol{u}}(\boldsymbol{x}) \geq h(\boldsymbol{x}) - \delta & \forall \boldsymbol{x} \in \overline{\mathcal{C} \setminus \mathcal{X}_r} \\ 0 \leq \delta < 1 \end{cases} \tag{27}$$

where $\boldsymbol{u}(\boldsymbol{x}) : \overline{\mathcal{C}} \to \mathcal{U}$ is a locally Lipschitz parameterized function with some unknown parameters, and c is a large constant that penalizes safety/reachability violations.

Proposition 4: Suppose $(\boldsymbol{u}^*(\boldsymbol{x}), \delta^*)$ is obtained by solving optimization (27). The controller $\boldsymbol{u}^*(\boldsymbol{x})$ is a reach-avoid controller with respect to the sets $\mathcal{D}$ and $\mathcal{X}_r$, where $\mathcal{D} = \{\, \boldsymbol{x} \mid h'(\boldsymbol{x}) > 0 \,\}$ with $h'(\boldsymbol{x}) = \frac{h(\boldsymbol{x})-\delta}{1-\delta}$.

IV. EXAMPLES

This section demonstrates the theoretical developments of the proposed methods numerically. The semidefinite programming tool Mosek [12] is employed to address the optimization involved. Some parameters are presented in Table I, and the computed $\|\boldsymbol{u}(\boldsymbol{x}) - \boldsymbol{k}(\boldsymbol{x})\|$'s are summarized in Tables II and III. In the following examples, we attempt to synthesize a reach-avoid controller of the linear form in a minimal way of modifying the controller $\boldsymbol{k}(\boldsymbol{x}) = \boldsymbol{0}$.

A. Deterministic Case

Example 2: Consider a simple mobile robot navigation that can be described by the following equation [15]:

$$\begin{cases} \dot{x} = u_1 \\ \dot{y} = u_2 \end{cases} \tag{28}$$

where $\mathcal{C} = \{(x,y)^\top \mid 1 - x^2 - y^2 > 0\}$, $\mathcal{X}_r = \{(x,y)^\top \mid 0.01(x-0.9)^2 + y^2 - 0.01 < 0\}$, and $\mathcal{U} = \{\, (u_1,u_2)^\top \mid 10^{-3} \leq u_1 \leq 1, -1 \leq u_2 \leq 1 \,\}$.

The vector fields formed by the computed controllers are visualized in Fig. 1 . It is concluded from Table II that the reach-avoid controller generated by exponential guidance-barrier functions can be improved by asymptotic and lax ones, and the lax guidance-barrier function generates the best controller.

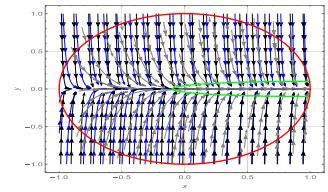

Fig. 1. Illustration of reach-avoid controllers for Example 2. Red and green curves denote the boundaries of safe set $\mathcal{C}$ and target set $\mathcal{X}_r$, respectively. Blue, black and gray curves, respectively, denote vector fields with controllers computed via solving optimization (4), (6), and (16).

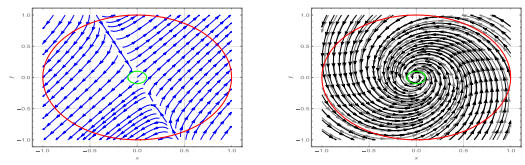

Fig. 2. Illustration of controller synthesis for Example 3. Red and green curves denote the boundaries of the sets $\mathcal{C}$ and $\mathcal{X}_r$, respectively. Left: The vector fields for Example 3 with the controller $\boldsymbol{k}(\boldsymbol{x})$. Right: Gray and black curves, respectively, denote the vector fields with controllers computed via solving optimization (4) and (6), respectively.

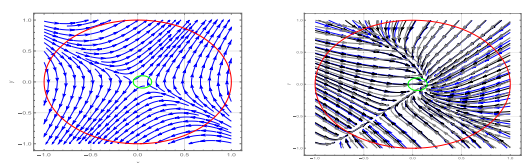

Fig. 3. Illustration of controller synthesis for Example 4. Red and green curves denote the boundaries of the sets $\mathcal{C}$ and $\mathcal{X}_r$, respectively. Left: The vector fields for Example 4 with the controller $\boldsymbol{k}(\boldsymbol{x})$. Right: Blue, gray and black curves, respectively, denote the vector fields with controllers computed via solving optimization (4), (6), and (16), respectively.

Example 3: Consider a second-order linear model from [17]

$$\begin{cases} \dot{x} = 2x + y + u_1 \\ \dot{y} = 3x + y + u_2 \end{cases} \tag{29}$$

where $\mathcal{C} = \{\, (x,y)^\top \mid 1 - x^2 - y^2 > 0 \,\}$, $\mathcal{X}_r = \{\, (x,y)^\top \mid x^2 + y^2 - 0.01 < 0 \,\}$, and $\mathcal{U} = \{\, (u_1,u_2)^\top \mid -3 \leq u_1 \leq 3, -3 \leq u_2 \leq 3 \,\}$.

Fig. 2 shows the vector fields created by the computed controllers. Based on the results in Table II, it is evident that the reach-avoid controller derived using exponential guidance-barrier functions can be enhanced by using either asymptotic or lax functions. However, the controller generated by the asymptotic guidance-barrier function cannot be further improved using the lax function.

Example 4: Consider a second-order polynomial nonlinear control affine system from [17]

$$\begin{cases} \dot{x} = y + (x^2 + y + 1)u_1 \\ \dot{y} = x + \frac{1}{3}x^3 + y + (y^2 + x + 1)u_2 \end{cases} \tag{30}$$

where $\mathcal{C} = \{(x,y)^\top \mid 1 - x^2 - y^2 > 0\}$, $\mathcal{X}_r = \{(x,y)^\top \mid (x-0.05)^2 + y^2 - 0.01 < 0\}$, and $\mathcal{U} = \{\, (u_1,u_2)^\top \mid -2 \leq u_1 \leq 2, -2 \leq u_2 \leq 2 \,\}$.

The vector fields driven by the computed controllers are showcased in Fig. 3 . The reach-avoid controller generated by exponential guidance-barrier functions is enhanced by asymptotic and lax

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ones, although the improvements are not significant, as illustrated Table II.

B. Stochastic Case

Example 5: Consider the following stochastic system adapted from Example 2

$$\begin{cases} \dot{x} = u_1 + dw \\ \dot{y} = u_2 \end{cases} \tag{31}$$

where $\mathcal{C} = \{ (x, y)^\top \mid 1 - x^2 - y^2 > 0 \}$, $\mathcal{X}_r = \{ (x, y)^\top \mid 0.01(x - 0.9)^2 + y^2 - 0.01 < 0 \}$, and $\mathcal{U} = \mathbb{R}^2$.

We synthesize a reach-avoid controller $\boldsymbol{u} = (u_1, u_2)^\top$. It is concluded from Table III that the reach-avoid controller generated by asymptotic guidance-barrier functions improve the one from exponential guidance-barrier functions.

Examples 3–5 demonstrate the effectiveness of exponential control guidance-barrier functions in generating reach-avoid controllers that perform comparably to those derived using asymptotic and lax control guidance-barrier functions, for certain systems. In addition, due to the convexity of optimization (4) over the controller $\boldsymbol{u}(\boldsymbol{x})$, it is recommended to use this approach to synthesize reach-avoid controllers in practical applications, provided that there is no $\boldsymbol{y} \in \mathcal{C} \setminus \mathcal{X}_r$ such that $h(\boldsymbol{y}) \geq \sup_{\boldsymbol{x} \in \mathcal{X}_r} h(\boldsymbol{x})$. Safe stabilization of a neighborhood $\mathcal{X}_r$ around an asymptotic equilibrium state is one scenario where it may not be possible to find a $\boldsymbol{y} \in \mathcal{C} \setminus \mathcal{X}_r$ that satisfies $h(\boldsymbol{y}) \geq \sup_{\boldsymbol{x} \in \mathcal{X}_r} h(\boldsymbol{x})$. Moreover, using the reach-avoid controller obtained from solving optimization (4) can expedite the iterative procedure (Algorithm 2) for solving optimizations (6) and (16), making it an excellent starting point for the process. We also use Algorithm 3 to solve optimizations (6) and (16), which results in the same outcomes as those in Tables II and III, but with higher computational costs. The only exception is optimization (16) for Example 4, where we did not obtain a reach-avoid controller. While optimizations (6) and (16) are nonconvex, they are subject to weaker constraints than (4) and can generate better reach-avoid controllers, especially for Example 2. Thus, the exponential, asymptotic, and lax control guidance-barrier functions have their unique strengths and drawbacks, but can work together in a complementary manner to improve the effectiveness of practical applications.

V. Conclusion

In this article, we investigated the problem of synthesizing reach-avoid controllers for deterministic systems modeled by ordinary differential equations and stochastic systems modeled by stochastic differential equations, based on the notion of control guidance-barrier functions. Three control guidance-barrier functions were proposed for deterministic systems, which provide increasingly weaker sufficient conditions for synthesizing reach-avoid controllers, and two of these functions were extended to stochastic systems. Finally, several numerical examples demonstrate the performance of proposed methods.

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